

Week 1

Foundations of Data Science: K-Means Clustering in Python

Week 1: Foundations of Data Science and K-Means Clustering

Introduce yourself, and tell us a bit about why you are doing this course -- and what you hope to gain from it.

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Week 1: Foundations of Data Science: K-Means Clustering in Python



This week we will introduce you to the course and to the team who will be guiding you through the course over the next 5 weeks. The aim of this week's material is to gently introduce you to Data Science through some real-world examples of where Data ...

[More](#)

Learning Objectives

- How Data Science might be applied to real-world situations.
- How the K-Means clustering algorithm works.
- What we mean by data.
- What we mean by machine learning.

Welcome and Introduction

Video: Welcome and Introduction 2 min

[Resume](#)



 **Video:** Introduction to Data Science 2 min

 **Discussion Prompt:** Welcome! 30 min

Data

 **Video:** What is Data? 1 min

 **Discussion Prompt:** Examples of Data 2h

 **Video:** Types of Data 1 min

 **Practice Quiz:** Types of Data – Review Information 2 questions

Machine Learning

 **Video:** Machine Learning 3 min

 **Discussion Prompt:** Machine Learning in the News 2h

Supervised vs Unsupervised Learning

 **Video:** Supervised vs Unsupervised Learning 2 min

 **Practice Quiz:** Supervised vs Unsupervised – Review Information 2 questions

Clustering with K-Means

 **Video:** K-Means Clustering 4 min





Practice Quiz: K-Means Clustering - Review Information

2 questions



Video: Preparing your Data 1 min

Week 1 Outro and Assessment



Video: A Real World Dataset 53 sec



Quiz: Week 1 Summative Assessment 5 questions